

WASP-10b

R-1126

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_201101 · created 2026-07-30T22:31:33+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459154.6787 +/- 0.0047 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.058
Transit depth (Rp/Rs)^2	2.35 +/- 1.78 [%]
Orbital Inclination (inc)	87.65 +/- 1.8
Airmass coefficient 1 (a1)	1.162 +/- 0.05
Airmass coefficient 2 (a2)	-0.15 +/- 0.19
Scatter in the residuals of the lightcurve fit is	4.31 %
Best Comparison Star	#3 - [330, 211]
Optimal Aperture	2.43
Optimal Annulus	12.11
Transit Duration (day)	0.08 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.058 vs 0.1592 (deviation 0.11σ)
Duration fit vs published	0.08 d vs 0.0946 d (deviation 0.73σ)
Mid-time O–C	-0.6 min
Depth SNR	2.6
Residual scatter	4.31 %

Light curve

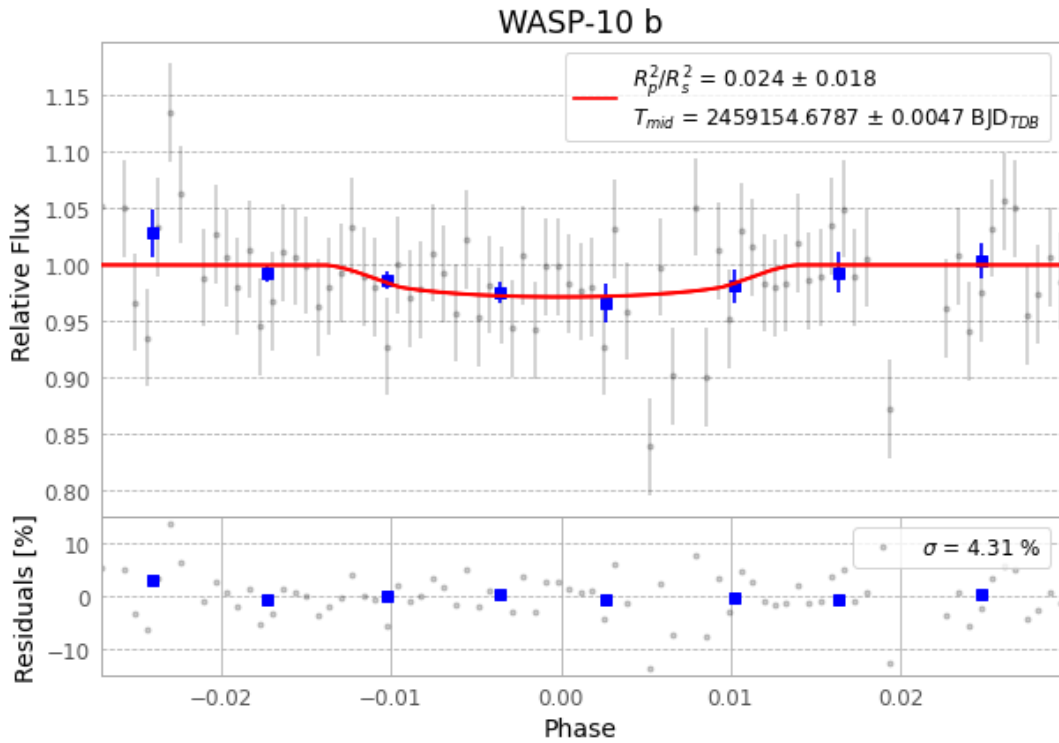


Figure 1 — FinalLightCurve_WASP-10 b_2020-10-31.png (SHA-256 cc4c890ee046fa84...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [332, 310], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e6204d0a5fff661e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

85 FITS frames (56 MB), WASP-10201101020912.FITS → WASP-10201101062112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-201101.log (7370b46e4304...), FinalLightCurve_WASP-10 b_2020-10-31.png (cc4c890ee046...), FinalLightCurve_WASP-10 b_2020-10-31.csv (6a3407e629cc...)

Compute resources

Wall time 280.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 22:31Z.